Supplementary Material: Directional Prompting for Gaze-Referential Inference in Large Multimodal Models

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This document provides more details excluded from the main paper due to space constraints: construction details for CG-GRI, the LMM-generated-arrow prompt, and an analysis of `Gemini-3.1-pro`'s sensitivity to UniGaze-predicted gaze arrows.

1 Construction of CG-GRI

CG-GRI (N=483) is derived from VideoAttentionTarget [1] (N=200) and GazeFollow [2] (N=283). Neither source provides GRI-ready candidate lists, so we construct the multiple-choice options and targets as follows. We refer to the two subsets as VAT-GRI and GF-GRI; they are reporting slices of one CG-GRI track rather than separate evaluation tracks.

VideoAttentionTarget (VAT-GRI). As consecutive frames within a clip are visually similar, we sample by gaze sequence or clip and retain at most one frame per clip. For each selected frame, automatic screening uses YOLOv10L [3]: the observer head box must be valid; detections overlapping the observer box are removed; the annotated gaze point must fall within at least one remaining box; and at least two candidate boxes must remain. Candidate descriptions are drafted by an LMM and human-edited so that instances of the same class remain visually distinguishable. Every retained instance is human-annotated for observer identity, candidate list, and the target indicated by the original VAT gaze point; ambiguous images are discarded. Ground-truth arrow origins are also human-annotated eye points, while endpoints remain the original VAT gaze points. This process yields 200 VAT-GRI instances.

GazeFollow (GF-GRI). GazeFollow provides web images with gaze-region annotations but no gazed-object labels. We randomly sample 1,000 images and retain those in which the observer's eyes and gaze direction can be judged. Images with occluded or unclear eyes are discarded. For every retained image, candidate objects are drafted using YOLO and human-edited, and the gazed target is human-annotated using the original GazeFollow heatmap. Ground-truth arrows use the original eye annotation as the origin and the original gaze point as the endpoint. This process yields 283 GF-GRI instances.

Table 1. Model outcomes on the Arrow-May-Help and Arrow-May-Harm sets of CG-GRI.

Diagnostic subset and outcome	`Gemini-3.1-pro`	`Qwen3.5-27B`
Arrow-May-Help: RGB+A(UniGaze) correct (N=90)	**87/90 (96.7%)**	71/90 (78.9%)
Arrow-May-Harm: RGB+A(UniGaze) wrong (N=83)	**70/83 (84.3%)**	48/83 (57.8%)

Shared protocol. The target observer is marked by a red unfilled bounding-box outline in every CG-GRI condition, including the RGB baseline. The box disambiguates the observer and selects whose gaze is predicted, but it is not used as the arrow origin. Among detected faces overlapping the box, we retain the prediction whose projected origin is closest to the reference head. The question lists the human-annotated candidate descriptions and asks which candidate the marked observer is looking at.

2 LMM-Generated-Arrow Prompt

The first stage of RGB+LMM-gen receives no answer choices and returns a gaze ray that is rendered on the image before the plain multiple-choice prompt is applied.

```
Identify the target person's gaze direction in this image.
Do not name an object.
Return exactly one JSON object using a normalized 0-1000 image coordinate system:
{"start": [x, y], "end": [x, y]}.
The start must be near the target person's visible eyes, and the end must lie in the
gaze direction.
```

The JSON `answer` field of the second-stage multiple-choice response is scored by case-insensitive exact match against the ground-truth label.

3 Sensitivity of Gemini-3.1-pro to UniGaze-Predicted Gaze Arrows

The main paper reports a significant accuracy degradation for `Gemini-3.1-pro` in the 3-candidate stratum of CG-GRI, where RGB+A(UniGaze) underperforms RGB, while other LMMs such as `Qwen3.5-27B` see RGB+A(UniGaze) outperforming RGB. Motivated by this result, we analyze the effect of A(UniGaze) on `Gemini-3.1-pro` and `Qwen3.5-27B`.

We first consider two sets of samples: (a) the *Arrow-May-Help* set, covering samples for which at least one of the two LMMs predicts wrongly with RGB but correctly with RGB+A(UniGaze); and (b) the *Arrow-May-Harm* set, covering samples for which at least one of the two LMMs predicts correctly with RGB but incorrectly with RGB+A(UniGaze). Table 1 shows that these sets contain 90 and 83 samples, respectively.

For the Arrow-May-Help set, Gemini yields 16 more correct predictions than Qwen (87 vs. 71). This suggests that when the arrow helps, Gemini can more

effectively benefit from it. The finding is quite different for the Arrow-May-Harm set. On this set, Gemini suffers 70 wrong predictions with RGB+A(UniGaze) out of the 83 samples that the arrow may harm, whereas Qwen suffers only 48 out of 83. In conclusion, Gemini is relatively more susceptible to arrow cues that may harm LMMs in CG-GRI tasks.

References

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